High Altitude Pulmonary Edema (HAPE)

Background

- High risk with travel to 8,000ft (2,500 m) or higher
- May occur at lower altitudes in susceptible population (COPD, CHF, obesity, history of AMS)
- Development of symptoms based on:
  - Rate of Ascent
  - Sleeping altitude
  - Strength of hypoxic ventilatory response
  - Alcohol intake
  - Obesity
- NOT based on physical fitness, age, sex, smoking, previous high-altitude experience
- Tend to have recurrence of symptoms whenever they return to the symptomatic altitude
- Low partial pressure of O2 leads to decreased diffusion of O2 across alveolar spaces --> hypoxemia results in tachypnea and cerebral vasodilation (↓ CO2) --> increased ICP

Definition

In the setting of a recent gain in altitude, the presence of the following:

- Symptoms: at least two of:
  - Dyspnea at rest
  - Cough
  - Weakness or decreased exercise performance
  - Chest tightness or congestion
- Signs: at least two of:
  - Crackles or wheezing in at least one lung field
  - Central cyanosis
  - Tachypnea
  - Tachycardia

EMT/AEMT

- Follow General Medical Care Guideline
- Perform a full neurologic and pulmonary exam
- **Descend at least 1000m or until symptoms resolve**
- If practical, Oxygen to improve SpO2 >90%

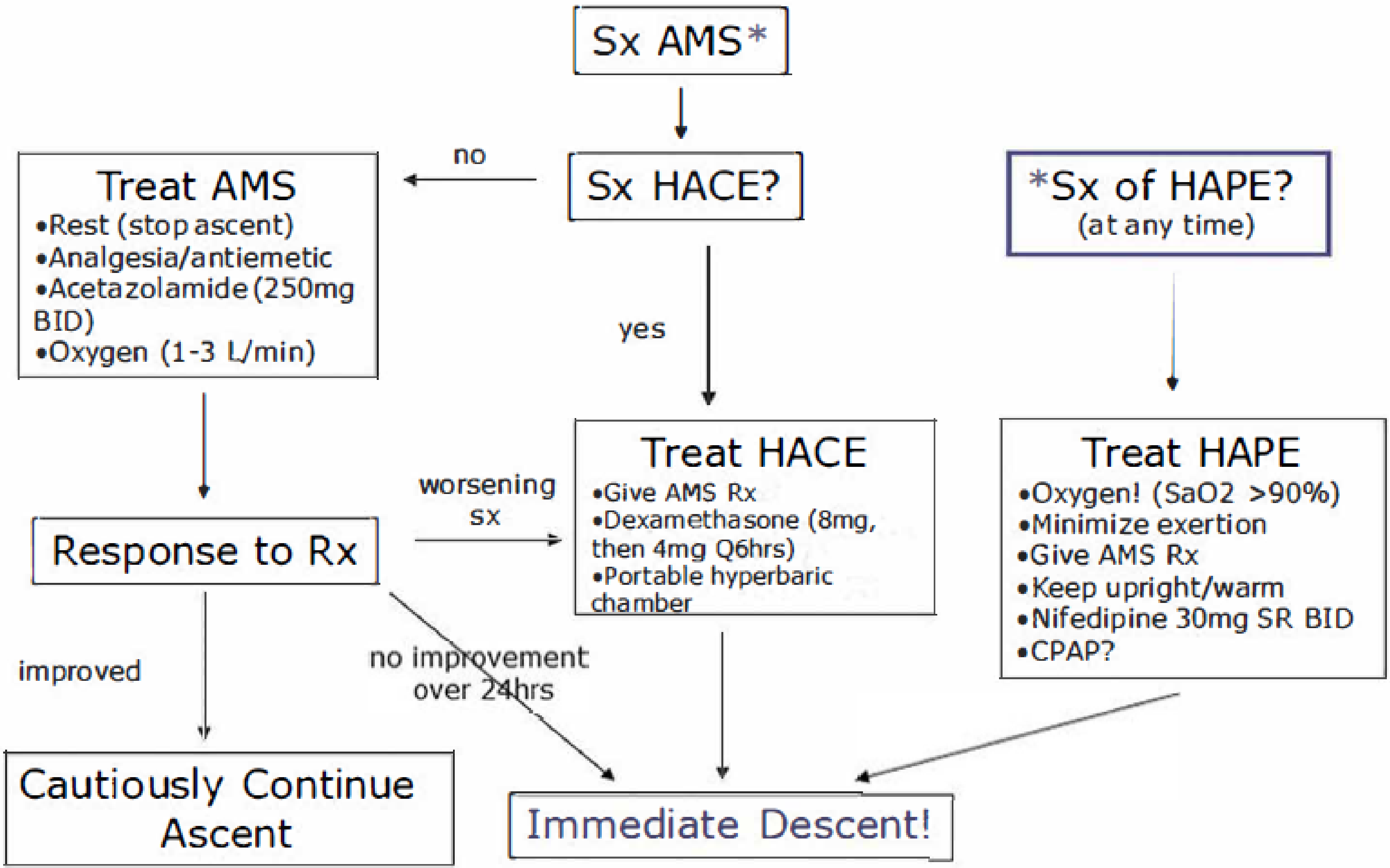
Paramedic

- Follow General Medical Care Guideline
- **Nifedipine slow release 30mg every 12 hours** - in addition to descent (if available)

After descent due to HAPE and symptoms have resolved, patient may attempt reascent
Patient should be evaluated by a physician after descent

Expected SpO_2 and PaO_2 levels at altitude[2]

Altitude	SpO_2	PaO_2 (mm Hg)
1,500 to 3,500 m (4,900 to 11,500 ft)	about 90%	55-75
3,500 to 5,500 m (11,500 to 18,000 ft)	75-85%	40-60
5,500 to 8,850 m (18,000 to 29,000 ft)	58-75%	28-40


Sx AMS*
Sx HACE?
no
yes
Treat AMS
•Rest (stop ascent)
•Analgesia/antiemetic
•Acetazolamide (250mg BID)
•Oxygen (1-3 L/min)
*Sx of HAPE?
(at any time)
Treat HACE
•Give AMS Rx
•Dexamethasone (8mg, then 4mg Q6hrs)
•Portable hyperbaric chamber
Treat HAPE
•Oxygen! (SaO2 >90%)
•Minimize exertion
•Give AMS Rx
•Keep upright/warm
•Nifedipine 30mg SR BID
•CPAP?
worsening sx
Response to Rx
improved
no improvement over 24hrs
Cautiously Continue Ascent
Immediate Descent!